

SatQuery AI - Analysis Report

Job smoke_1787991785395 | task: change_vqa

Query

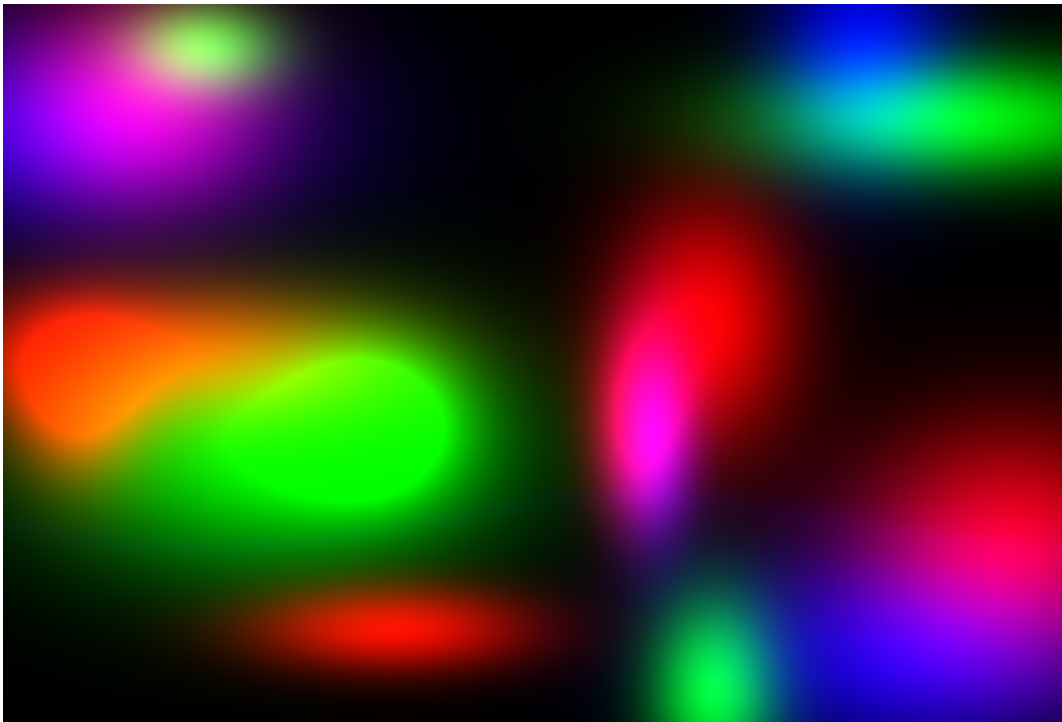
Has the built-up area increased, decreased, or remained unchanged?

Answer

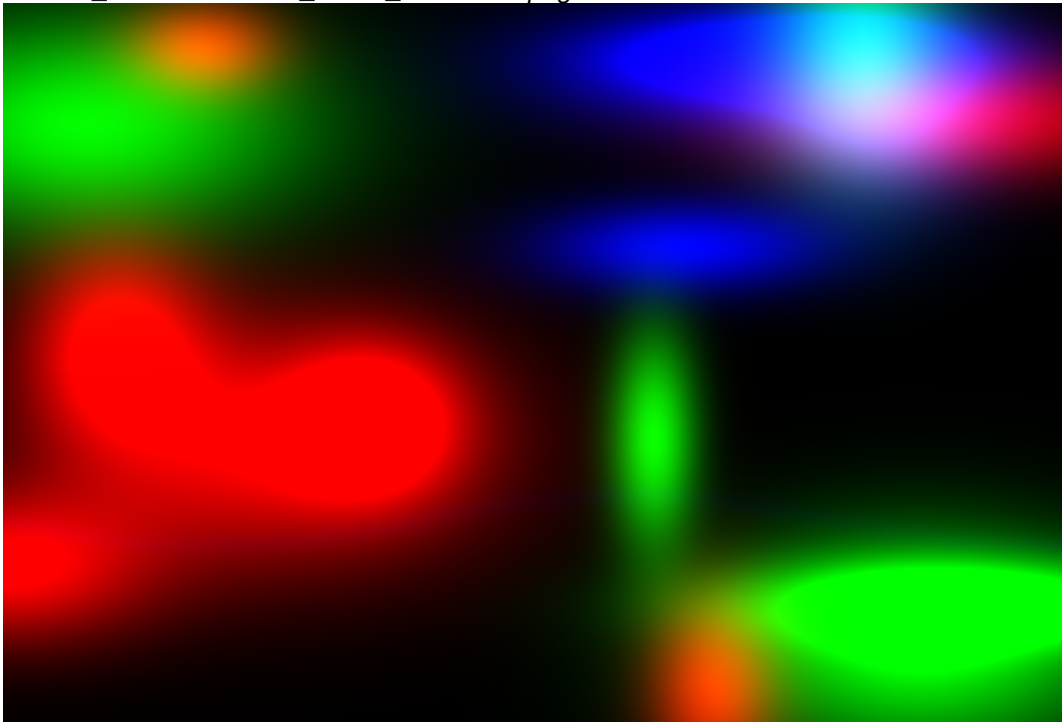
Based on the bi-temporal pair (2023-01-01 -> 2023-07-01), the built-up area has increased. [mock output - wired to real specialist once adapters ship]

Confidence

mock — Deterministic placeholder (skeleton/demo mode), not a calibrated score.



Evidence 1: smoke_1787991785395_areaA_20230101.png



Evidence 2: smoke_1787991785395_areaA_20230701.png

Audit trace (evaluated artifact)

```
{ "schema_version": "1.0.0", "task": "change_vqa", "query": "Has the built-up area increased, decreased, or remained unchanged?", "query_hash": "afe989e794", "input_config": { "n_images": 2, "images": [ { "filename": "C:\\project\\satquery\\backend\\runtime\\demo_data\\areaA_20230101.tif", "format": "GEOTIFF", "modality": "optical", "width": 512, "height": 512, "band_count": 3, "dtype": "uint8", "crs": "EPSG:4326", "extent": [ 72.8, 19.0, 72.84, 19.04 ], "pixel_size": [ 7.812500000001221e-05, 7.812499999999833e-05 ], "acquisition_date": "2023-01-01", "metadata_source": "none" }, { "filename": "C:\\project\\satquery\\backend\\runtime\\demo_data\\areaA_20230701.tif", "format": "GEOTIFF", "modality": "optical", "width": 512, "height": 512, "band_count": 3, "dtype": "uint8", "crs": "EPSG:4326", "extent": [ 72.8, 19.0, 72.84, 19.04 ], "pixel_size": [ 7.812500000001221e-05, 7.812499999999833e-05 ], "acquisition_date": "2023-07-01", "metadata_source": "none" } ], "geolocated": true, "co_registered": true, "co_registration_note": "CRS=EPSG:4326, extent overlap 100%", "bi_temporal": true, "dates_differ": true, "date_note": "Dates 2023-01-01 -> 2023-07-01", "modality_signature": [ "optical", "optical" ] }, "classification": { "task": "change_vqa", "method": "rules", "reasons": [ "change intent + interrogative structure" ], "alternatives": { "single_vqa": 0.0, "single_caption": 0.0, "single_grounding": 0.0, "change_vqa": 1.0, "change_description": 0.0, "sar_optical_fusion": 0.0 } }, "validation": { "passed": [ "format:GEOTIFF", "format:GEOTIFF", "temporal_dates_differ", "GEOTIFF (512x512, 3 band, uint8)", "GEOTIFF (512x512, 3 band, uint8)", "Bi-temporal/mono-modal pair detected.", "CRS=EPSG:4326, extent overlap 100%", "Dates 2023-01-01 -> 2023-07-01" ], "warnings": [], "failed": [], "checks": [ { "name": "format", "status": "passed", "message": "GEOTIFF (512x512, 3 band, uint8)" }, { "name": "format", "status": "passed", "message": "GEOTIFF (512x512, 3 band, uint8)" }, { "name": "pair_type", "status": "passed", "message": "Bi-temporal/mono-modal pair detected." }, { "name": "co_registration", "status": "passed", "message": "CRS=EPSG:4326, extent overlap 100%" }, { "name": "temporal", "status": "passed", "message": "Dates 2023-01-01 -> 2023-07-01" } ], "input_config": { "n_images": 2, "images": [ { "filename": "C:\\project\\satquery\\backend\\runtime\\demo_data\\areaA_20230101.tif", "format": "GEOTIFF", "modality": "optical", "width": 512, "height": 512, "band_count": 3, "dtype": "uint8", "crs": "EPSG:4326", "extent": [ 72.8, 19.0, 72.84, 19.04 ], "pixel_size": [ 7.812500000001221e-05, 7.812499999999833e-05 ], "acquisition_date": "2023-01-01", "metadata_source": "none" }, { "filename": "C:\\project\\satquery\\backend\\runtime\\demo_data\\areaA_20230701.tif", "format": "GEOTIFF", "modality": "optical", "width": 512, "height": 512, "band_count": 3, "dtype": "uint8", "crs": "EPSG:4326", "extent": [ 72.8, 19.0, 72.84, 19.04 ], "pixel_size": [ 7.812500000001221e-05, 7.812499999999833e-05 ], "acquisition_date": "2023-07-01", "metadata_source": "none" } ], "geolocated": true, "co_registered": true, "co_registration_note": "CRS=EPSG:4326, extent overlap 100%", "bi_temporal": true, "dates_differ": true, "date_note": "Dates 2023-01-01 -> 2023-07-01", "modality_signature": [ "optical", "optical" ] }, "accepted": true }, "registry_entries_used": [ { "id": "change_vqa", "task": "change_vqa", "model_id": "Qwen/Qwen3-VL-4B-Instruct", "adapter": null, "quantization": null, "fallback": "mock", "permitted_parameters": { "max_new_tokens": 512, "temperature": 0.2 } } ], "execution_time_ms": 60, "confidence": { "value": 0.8, "source": "mock", "note": "Deterministic placeholder (skeleton/demo mode), not a calibrated score." }, "outputs": { "text": { "text": "Based on the bi-temporal pair (2023-01-01 -> 2023-07-01), the built-up area has increased. [mock output - wired to real specialist once adapters ship]", "n_shots": 1 }, "boxes": [], "mask": { "path": null, "kind": "none" }, "evidence_thumbnails": [ "C:\\project\\satquery\\backend\\runtime\\outputs\\smoke_1787991785395_areaA_20230101.png", "C:\\project\\satquery\\backend\\runtime\\outputs\\smoke_1787991785395_areaA_20230701.png" ] }, "model_backend": "mock", "created_at": "2026-08-29T08:23:05.455621+00:00", "server_version": "0.1.0" }
```